

Review of Statistics 1 Material

A quick review of linear modeling

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Learning Objectives

Here, we will review functions and statistical methods you learned in Statistics 1. Specifically, we will review how to summarize and visualize numeric data and how to run a simple linear model.

New Functions

- `lm()` Runs a linear regression
- `ggplot()` Creates a plot

Set Up

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.0      v readr      2.2.0
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2    4.0.2      v tibble    3.3.1
## v lubridate  1.9.5      v tidyr     1.3.2
## v purrr      1.2.1
```

```
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
porsche <- read_csv("~/Sds_101A_SSI26/Class/Code/Workshop_Day3/PorschePrice.csv")
```

```
## Rows: 30 Columns: 3
```

```
## -- Column specification -----
```

```
## Delimiter: ","
```

```
## dbl (3): Price, Age, Mileage
```

```
##
```

```
## i Use 'spec()' to retrieve the full column specification for this data.
```

```
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

Before Class:

Review Chapters 3-22 in your textbook from Statistics 1.

During Class

Example 1.1: Porsche price vs. mileage

First, let's examine our data. We are interested in how the price of a porsche changes based on the mileage.

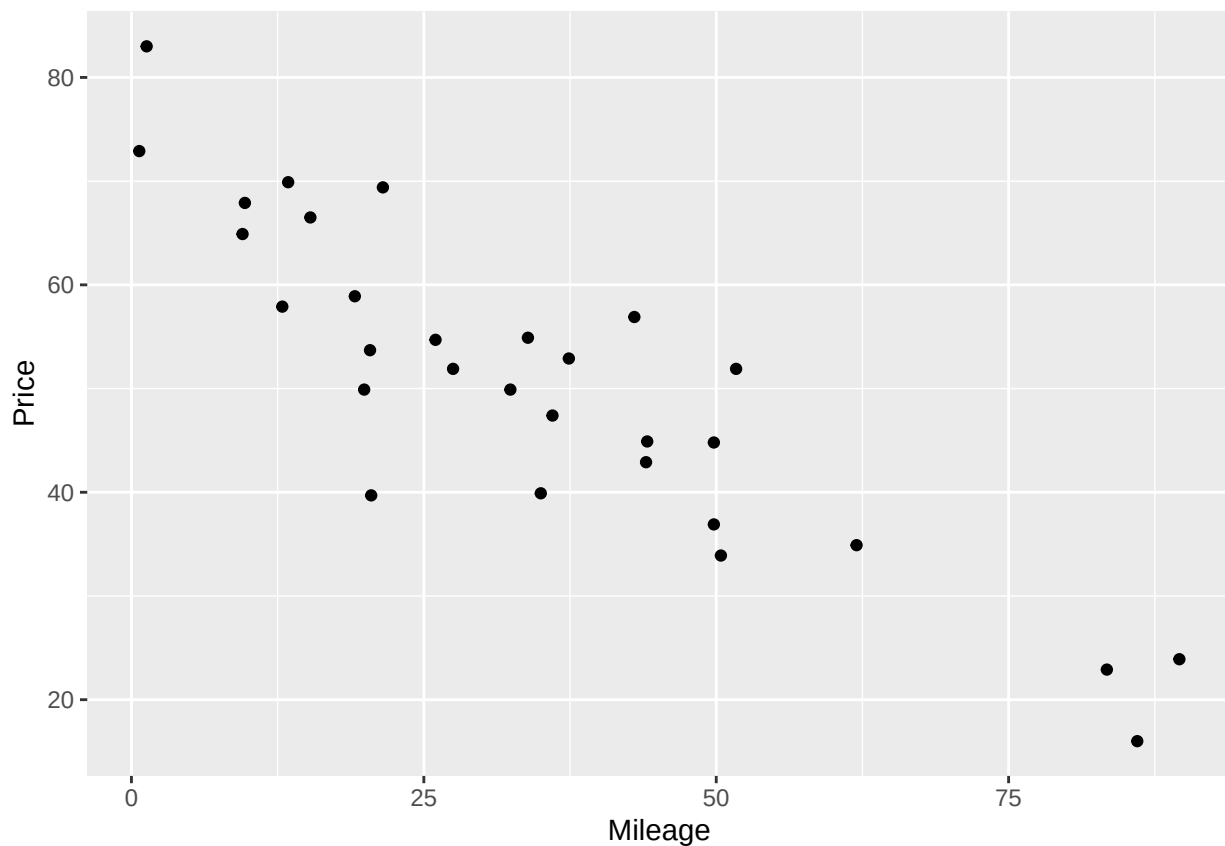
```
#Create a summary of the three variables
```

```
summary(porsche)
```

```
##      Price      Age      Mileage
##  Min.   :16.00  Min.   : 0.00  Min.   : 0.67
## 1st Qu.:40.65  1st Qu.: 3.00  1st Qu.:19.30
##  Median:51.90  Median: 4.00  Median:33.15
##   Mean  :50.54   Mean   : 6.20   Mean   :34.87
## 3rd Qu.:58.65  3rd Qu.: 7.75  3rd Qu.:48.38
##   Max.   :83.00   Max.   :22.00   Max.   :89.60
```

```
#create a scatterplot showing the relationship between mileage and price
```

```
ggplot(data = porsche, aes(x = Mileage, y = Price))+
  geom_point()
```



Next, let's run a simple linear model:

```
porsche_fit <- lm(data = porsche, Price ~ Mileage)
```

```
summary(porsche_fit)
```

```
##
```

```
## Call:
## lm(formula = Price ~ Mileage, data = porsche)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -19.3077  -4.0470  -0.3945   3.8374  12.6758
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  71.09045    2.36986   30.0 < 2e-16 ***
## Mileage      -0.58940    0.05665  -10.4 3.98e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.17 on 28 degrees of freedom
## Multiple R-squared:  0.7945, Adjusted R-squared:  0.7872
## F-statistic: 108.3 on 1 and 28 DF,  p-value: 3.982e-11
```

1. What is the slope estimate? Give an interpretation of the slope in the context of the question.
2. What is the R-squared value? Give an interpretation of the R-squared value in the context of the question.

Because we only have a single slope and intercept term, we can easily use `geom_smooth()` to add the line of best fit:

```
ggplot(data = porsche, aes(x = Mileage, y = Price))+
  geom_point() +
  geom_smooth(method = "lm", se = FALSE)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

